

Voluntarily-generated unimanual preparation is associated with stopping success: evidence from LRP and lateralized mu ERD before the stop signal

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Abstract According to the race models of the stop-signal paradigm, stopping success (successful vs. unsuccessful stopping) is attributed to the finishing times of a go and a stop process. In addition to those factors involving processing times, in the present study we sought to use electrophysiological measures to find factors involving activations that could affect stopping success. We hypothesized that voluntarily-generated unimanual preparation would be a factor. To assess voluntarily-generated unimanual preparation in the stop-signal paradigm, we used a selective-stopping task without any precue. The selective-stopping task also allowed us to assess reaction times (RTs) even when stopping was successful. We demonstrated shorter RTs in signal-respond (i.e. unsuccessful stopping) than in signal-inhibit (successful stopping) trials, as is predicted by the race models. More importantly, we also demonstrated different pre-signal lateralized readiness potentials between the two types of trials and larger lateralized mu ERD in signal-respond than in signal-inhibit trials, suggesting that voluntarily-generated unimanual preparation affects stopping success. In addition to what is described in the race models of the stop-signal paradigm, the present results therefore demonstrated measures of pre-signal activations that could influence stopping success.

Response inhibition and stopping success

Executive control is a critical ability in the ever-changing environment, and response inhibition has been thought to be

a primary component of executive control (e.g. Andres, 2003; Miyake, Friedman, Emerson, Witzki & Howerter, 2000). Response inhibition has been substantially studied with the stop-signal paradigm (van Boxtel, van der Molen, Jennings & Brunia, 2001; Verbruggen & Logan, 2008). In the stop-signal paradigm, an imperative or go stimulus is presented in every trial to indicate a response (for example, responding with either the left or the right hand according to the stimulus). Occasionally, a stop signal appears after the go stimulus with a short delay (the stop-signal delay or SSD) to indicate withholding of the planned response. In trials with the appearance of the stop signal (i.e. stop-signal trials), stopping of the planned or prepared response can either be successful or unsuccessful, which are referred to as signal-inhibit trials and signal-respond trials, respectively.

The independent horse race model of the stop-signal paradigm (Logan & Cowan, 1984) is widely used in accounting for stopping success (i.e. signal inhibit vs. signal respond). According to the independent race model, stopping success is determined by the relative finishing times of two independent processes: a go process initiated by the go stimulus and a stop process initiated by the stop signal. When the go process finishes first, the prepared response is executed; whereas when the stop process finishes first, the prepared response is stopped. The go process begins at the stimulus onset and its latency is the reaction time (RT); whereas the stop process begins at the stop-signal onset and its latency is the stop-signal reaction time or SSRT. Unlike RT, SSRT is unobservable, and a major merit of the independent race model is that it allows one to estimate SSRT. The method of estimating SSRT based on the independent race model is still valid even when the go and the stop processes are not entirely independent (Boucher, Palmeri, Logan & Schall, 2007). In short, relative to the stimulus onset, the finishing time of the go process is

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RT, and the finishing time of the stop process is SSRT plus the SSD. Stopping success is therefore affected by RT, SSRT, and SSD.

The influence of SSD is straightforward: when SSD is larger, the stop process is initiated later, resulting in a lower probability to win the race and succeed in stopping. The influence of SSD on stopping success is characterized in the inhibition functions (Logan & Cowan, 1984; Verbruggen & Logan, 2009a). In the race models (Boucher et al., 2007; Logan & Cowan, 1984), the latency of the stop process (i.e. SSRT) is a random variable in theory, but it is most frequently treated as a fixed value in practice since Logan & Cowan, (1984) showed that assuming SSRT as a constant is a good first approximation. The latency of the go process, which is estimated by RTs, is a random variable and described in a RT distribution of no-stop-signal trials, according to the race models. In addition to those factors involving processing times, could we find any factor involving activations that might also affect stopping success?

Voluntarily-generated unimanual preparation and performance

Different lines of evidence have suggested that prestimulus response preparation, which is response preparation that occurs before the imperative stimulus, could affect performance in RT tasks and stop-signal tasks. For example, by measuring the lateralized readiness potential (LRP; e.g. Coles, 1989; Mordkoff & Grosjean, 2001) from incompatible trials in a flanker task (Eriksen & Eriksen, 1974), Gratton, Coles, Sirevaag, Eriksen & Donchin, (1988) found that the lateralization *before* the onset of the imperative stimulus (indicating which hand should be used for responding) was larger in fast-RT than in slow-RT trials. Also, the direction of the prestimulus lateralization in fast-RT trials was associated with response accuracy. Those results suggest that prestimulus unimanual response preparation could affect RT and response accuracy. Moreover, with stop-signal tasks, Ko, Alsford & Miller, (2012) found that baseline response force was larger in signal-respond than in signal-inhibit trials, suggesting that degree of response tendency before the stimulus contributed to stopping success. Note that in those studies prestimulus response preparation was *voluntarily* generated in the sense that it was not reactive to a prestimulus cue (i.e. precue) conveying information about forthcoming responding or stopping. Also, because a unimanual response was required, prestimulus response preparation in those studies was unimanual preparation. Therefore, in the present study, we referred to this type of preparation as voluntarily-generated unimanual preparation. Voluntarily-generated

unimanual preparation is theoretically important because race models of the stop-signal paradigm are based on tasks without any precue.

Effects of unimanual preparation that is reactive to a precue have also been studied. For example, in a choice RT task, Jentzsch et al. (2004) found larger LRP amplitude (more negative LRP) when the precue specified which hand to respond.¹ With a go/nogo task, van den Wildenberg et al. (2002) compared rates of correct withholding between two types of episode. In contrast to the control episode, in the test episode, a precue was presented to inform participants that a nogo trial was imminent, which would result in smaller response readiness/tendency in the test episode. They found larger rate of correct withholding in the test episode when a nogo stimulus appeared, suggesting that smaller prestimulus response tendency in the test episode led to easier withholding. Moreover, Cai, Oldenkamp & Aron, (2011) measured motor evoked potentials (MEPs) from the right hand and compared them in a foreknowledge condition and a null condition. In the foreknowledge condition, a precue was given to indicate which hand would be stopped if the stop signal appeared (e.g. Maybe Stop Right); while in the null condition, “Null” was given as a precue, and no stimulus was shown and no response was required. They found smaller prestimulus MEP from the right hand in trials with the precue “Maybe Stop Right” than in the null trials, suggesting that the stop process had been initiated before the stop signal appeared. In addition, using a precue to indicate whether the stop signal was relevant, Verbruggen and Logan (2009b) found that participants responded slower when the precue indicated that the stop signal was relevant (i.e. stopping was needed when the stop signal appeared) than when the stop signal was irrelevant (no stopping was needed). Their results also indicated that larger RTs resulted from a raising response threshold, suggesting that prestimulus response preparation affected the go process and thus stopping performance. Moreover, in the end of their conclusion, they suggested that processes involved in tasks with or without precues might not be the same.

Nonetheless, whether voluntarily-generated unimanual preparation (e.g. prestimulus unimanual preparation *not* elicited by a precue in two-choice RT tasks) could affect stopping success is still not clear. Firstly, influences of voluntarily-generated unimanual preparation on performance reported in Gratton et al., (1988) did not consider performance in the stop-signal paradigm. Secondly, in Claffey, Sheldon, Stinear, Verbruggen & Aron, (2010) or

¹ In tasks without a precue, prestimulus LRPs most likely reflect the participant's guessing about responding hand (e.g. Gratton et al., 1988). In contrast, in tasks with a precue which specifies which hand to respond (e.g. Jentzsch et al., 2004), prestimulus LRPs reflect the advance preparation based on the precue.

Cai et al., (2011), influences of prestimulus unimanual preparation on stopping success (i.e. signal inhibit vs. signal respond) were not considered. In addition, in those studies prestimulus unimanual preparation was initiated in response to a precue rather than voluntarily or spontaneously produced as it is in standard stop-signal tasks or choice RT tasks; and processes in tasks with and without a precue might not be the same, as was suggested by Verbruggen and Logan (2009b). More importantly, because the race models (Boucher et al., 2007; Logan & Cowan, 1984), which are widely used to account for stopping performance, are based on tasks without using any precue (i.e. voluntarily-generated unimanual preparation is relevant). Therefore, it would be useful to investigate influences of voluntarily-generated unimanual preparation on stopping performance.

Selective response inhibition and selective-stopping tasks

As has been noted, most stop-signal tasks examine inhibiting a single activated response, because there is only one response activated in a single trial (Verbruggen & Logan, 2008). However, we normally possess several concurrently activated response tendencies (rather than one) when we are awake, and in most cases we need to stop a subset of them rather than all to carry out our intention or respond to the environment. This ability has been referred to as selective response inhibition and studied using selective-stopping tasks (e.g. Aron & Verbruggen, 2008; Cai et al., 2011; Ko & Miller, 2011). In each trial of those tasks, a bimanual response was activated in response to the go stimulus. This would be especially true when there is *no* precue conveying which hand may be stopped (e.g. Coxon et al., 2007; Ko & Miller, 2011, 2013). In some trials, stopping was then signalled, indicating which hand to be stopped.

In comparison, in a stop-all task, only a single-hand response is activated, and it is to be stopped if the stop signal appears. In some stop-all tasks, stopping is required only when the stop signal appears after a designated stimulus (e.g. De Jong et al., 1995; Ko et al., 2012), which is usually also referred to as selective-stopping tasks. However, note that in those tasks, only one response is activated, and therefore, they are actually stop-all tasks in that regard.² For this reason, in the present study, the term “selective-stopping task” is reserved to refer to those tasks in which two concurrently activated response tendencies

are generated and in some trials one response is subsequently stopped. Importantly, we used the selective-stopping task in the present study, so that RTs could still be assessed from the hand not-to-be-stopped (i.e. the nontarget hand) even when stopping was successful.

Voluntarily-generated unimanual preparation in the selective-stopping task

The unimanual preparation *before the stop signal* in the selective-stopping task and that in a stop-all task are different in whether the preparation is generated voluntarily. In the selective-stopping task (without precues), any unimanual preparation before the stop signal is voluntarily generated, for the participant who does not know which hand to respond/stop before the stop signal appears. In contrast, in stop-all tasks, the go stimulus indicates which hand to respond, and therefore any unimanual preparation before the stop signal is in response to the go stimulus and is therefore not voluntarily generated. In the present study, we were concerned with voluntarily-generated unimanual preparation in the stop-signal paradigm that is conceptually similar to the prestimulus unimanual preparation in typical two-choice RT tasks without any precue (e.g. Gratton et al., 1988). For that purpose, in the present study, we used a selective-stopping task rather than a stop-all task and examined unimanual preparation in a pre-signal (i.e. before the stop signal) time interval.³

To assess unimanual preparation, the first measure that we employed was the lateralized readiness potential (LRP) derived from the event-related potentials, because the LRP has been routinely used to indicate unimanual preparation in the study of human information processing (e.g. Coles, 1989; Mordkoff & Grosjean, 2001). The second measure that we employed will be described next.

Mu rhythm ERD and movement preparation

As early as human electroencephalography (EEG) was invented, rhythms in the alpha frequency band ($\sim 8\text{--}12$ Hz) were found (Berger, 1929–1935 cited by

² Ko & Miller, (2011) have described the difference between the selective-stopping task that is used to assess our ability to selectively inhibit a subset of concurrently activated response tendencies and the selective-stopping task that is about our ability to distinguish what stimulus is subject to be stopped.

³ In standard two-choice RT tasks (requiring either a left-hand or right-hand response) without any precue, the participant does not know which hand should be prepared to respond until the appearance of the imperative stimulus. By the same token, in the selective-stopping task without any precue indicating which hand may be stopped, the participant does not know which hand should be prepared to stop until the appearance of the stop signal. Therefore, like response preparation before the imperative stimulus in the standard two-choice RT task, response preparation before the stop signal in the selective-stopping task without any precue is response preparation voluntarily generated.

Adrian, 1936). Rhythms in the alpha frequency band from EEG were then observed not only from occipital but also from precentral cortical areas (Jasper & Andrews, 1938). Subsequent studies showed that alpha rhythms from the Rolandic (precentral) area were modulated by movement rather than eyes' closing, indicating that alpha rhythms from occipital and Rolandic areas were dissociable, and the later was therefore separately denoted as Rolandic wicket rhythm or mu rhythm (e.g. Chatrian, Petersen & Lazarte, 1959; Gastaut, Naquet & Gastaut, 1965).

The power of EEG rhythms of a frequency band can either be decreased or increased by association with an event (e.g. hand movement or stimulus presentation), which has been referred to as an event-related desynchronization (ERD; Pfurtscheller & Aranibar, 1977) and an event-related synchronization (ERS; Pfurtscheller, 1992), respectively. ERD of mu rhythms (mu ERD) from over the sensorimotor cortex have been found to occur before and during voluntary hand movement or response (Pfurtscheller & Aranibar, 1979; Toro et al., 1994), and the effects tend to be largest on the side contralateral to the responding hand (Pfurtscheller & Berghold, 1989). It is worth noting that mu ERD before and during movement is consistently observed when movement is self-paced rather than reactive to an imperative stimulus or a precue conveying information about forthcoming movement (Pfurtscheller & Aranibar, 1977; Pfurtscheller & Berghold, 1989; Toro et al., 1994). In addition to self-paced movement, mu ERD has also been used to examine prestimulus response preparation in reaction-time tasks. For example, Deiber et al., (2012) found comparable mu ERD before the imperative stimulus when the response hand was freely chosen and when it was indicated by a precue. Therefore, mu ERD could be used to indicate voluntarily generated response preparation. Moreover, to assess unimanual response preparation, mu ERD and more generally, power changes, were calculated by the same subtraction method that is used in calculating the LRP (De Jong, Gladwin & Hart, 2006; Deiber et al., 2012; Doyle, Yarrow & Brown, 2005; Gladwin, Hart & De Jong, 2008), referred to as a lateralized mu ERD.

In short, in the present study, we aimed to test influences of voluntarily-generated unimanual preparation on stopping success, by comparing the LRPs with lateralized mu ERDs before the stop signal between signal-inhibit and signal-respond trials in a selective-stopping task without any precue.

Method

Participants

Eighteen students from National Central University participated in this experiment. One participant did not follow

the instruction of trying to stop in the stop-signal trials, resulting in an extremely small mean SSD (31 vs. 273 ms in the remaining 17 participants); that participant was therefore excluded. Participants were 4 female and 13 male, ranging in age from 19 to 27 years (mean = 21.6 - years). All had normal or corrected-to-normal vision. Excluding setting up for EEG recording, each participated in a single session lasting approximately 50 min and was paid NTD 500.

Stimuli

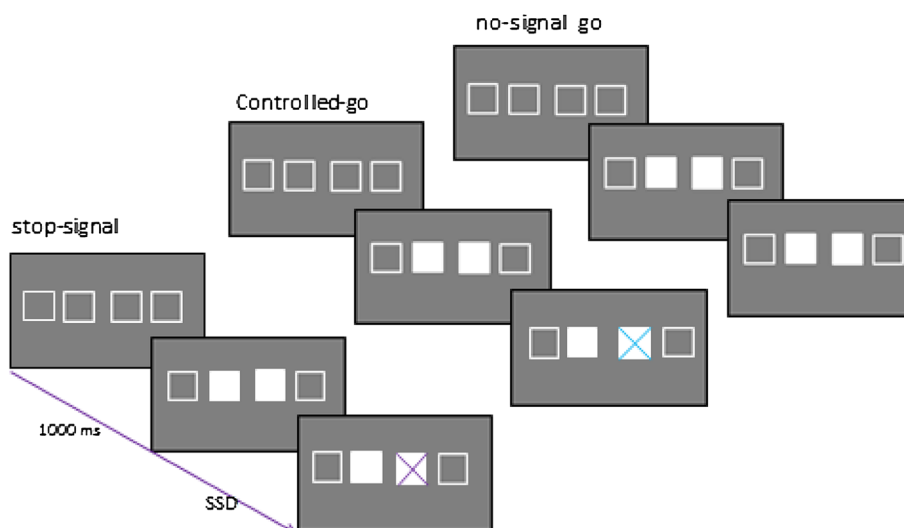
Four outline squares were deployed along the central horizontal line of the screen, and the length of any one side of each square was about 0.88° . The four outline squares were equally spaced with an inter-square distance of about 0.71° edge to edge; two of them were deployed on the left side of the vertical midline of the screen and the other two were on the right side of it. Therefore, with reference to the vertical midline, two squares are referred to as inner squares (the two that were closer to the vertical midline) and the other two as outer squares.

In each trial, either the two inner or the two outer outline squares were simultaneously filled in, and the two filled-in squares served as the primary-task (i.e. go) stimulus. The probabilities of presenting the two filled-in inner and the two filled-in outer squares were both 0.5. After a short delay, an X was sometimes shown on the diagonals of the left or the right filled-in square (equal probability for each side): an X was present in all stop-signal trials (25 % of all trials) and some go trials (25 % of all trials). Two different colours of X were used for each participant (purple or blue). The purple X served as the stop signal for half of the participants, with the blue X indicating not stopping, whereas these assignments were reversed for the other half. Those go trials with an X presented are referred to as controlled-go trials, which are designed to control for the stop signal as an extra stimulus (Ko & Miller, 2013). No X was shown in the remaining go trials (50 % of all trials), referred to as no-signal go trials. Figure 1 shows an instance in which the two inner squares are filled-in, and the purple X indicates stopping and appears on the right square.

Procedure

In each trial, participants were first presented with the four outline squares. After 1,000 ms, either the two inner squares or the two outer squares filled in (Fig. 1). If the two inner squares filled in, participants were instructed to simultaneously and quickly press the keys with their left and right index fingers; if the two outer squares filled in,

Fig. 1 A schematic diagram of the stimuli and the order of presentation. In this instance, the two inner squares are filled-in, and the purple X indicates stopping and appears on the right square



they were required to simultaneously and quickly press the keys with their left and right middle fingers.

In stop-signal and controlled-go trials, an X was presented after a short delay from the onset of the two filled-in squares. The X was superimposed on one of the two filled-in squares. Participants were instructed to stop their left-hand response while still quickly executing their right-hand response when the blue (or purple) X was presented on the left square; whereas they were to quickly execute the two hand response without stopping when the purple (or blue) X was presented on the square. These response requirements were symmetrically applied to trials in which the X was presented on the right square.

The delay from go stimulus onset to the appearance of the X in stop-signal trials (i.e. the SSD) was controlled by a staircase, starting at 300 ms and changing dynamically according to the performance in the stop signal trials. The delay in the next stop-signal trial increased 25 ms after each successful stop and decreased 25 ms after each unsuccessful stop. When the X in the controlled-go trials was presented, it was presented with the current value of the staircase, and the value of that staircase was not affected by performance in that trial.

Participants were instructed to make their responses immediately as soon as they saw the squares yet try to stop when an X of one indicated colour appeared. They were also informed that it was normal that they sometimes could not successfully stop when the X signalling stopping appeared. In addition, they were told that feedback would be given to help them follow the instructions. If they could not successfully stop when the stop signal appeared, they would get feedback saying ‘Try to stop the side with blue (or purple) X’. If their response was too slow (slower than individual’s mean RT plus one standard deviation in the

practice block or 800 ms, whichever was smaller), they would get feedback to warn them saying ‘You should respond faster!’. If they pressed more than two keys, they would get feedback saying ‘You pressed too many keys’. Finally, if they only made one response in controlled-go trials, they would get feedback saying ‘You should execute the side with purple (or blue) X’.

A single session consisted of a practice block with 64 trials and 5 experimental blocks with 128 trials in each. Breaks for rest were provided between blocks.

EEG data acquisition

EEG was continuously recorded during an experimental session using a 32-channel elastic cap, Scan 4.3 software and SynAmps amplifier system (Neuroscan, El Paso, TX, USA). All channels were online referenced to linked mastoids, and an electrode placed on the forehead anterior to the Fz electrode served as the ground electrode. Vertical EOG was monitored in a bipolar fashion from two electrodes placed approximately 2 cm above and below the left eye, and horizontal EOG was also monitored from bipolar electrodes placed about 2 cm lateral to the left and right outer canthi. Data were sampled every 1 ms (i.e. 1,000 Hz sampling rate), and all electrode impedances were kept below 5 k Ω .

ERP analyses and the LRP

Epoching was performed on EEGs recorded continuously during the experiment. The length of the epoch was determined by two steps: (1) epoch length of each participant was equal to 1,000 ms duration between the trial onset (the four outline squares) and the go-stimulus onset

(two filled-in squares) plus the maximum SSD of that participant obtained from the tracking procedure plus a 1,000-ms response window plus the maximum inter-trial interval (500 ms), (2) epoch length was equal to the minimum of the epoch lengths of all participants (so that averaging across participants could be performed), which was 2,800 ms in the present study. For each epoch, the 1,000-ms duration from the trial onset to the go-stimulus onset served as baseline duration for performing baseline correction. Because the X, either signalling stopping or not, was the critical event in the task, EEGs were time locked to the onset of the X and lined up to 200 ms time point after the go-stimulus onset except the no-signal go trials in which there were no X (those no-signal go trials were also excluded in further analyses). The X-locked EEGs in each trial were then sorted into conditions of interest and the X side. The LRPs were computed using those EEGs from the C3 and C4 sites, and the subtraction was performed according to the X side.⁴

Time–frequency analyses and the lateralized mu ERD

Time–frequency decompositions of EEG signals and the calculation of event-related spectrum perturbation or ERSP were carried out using the newtimef command in EEGLAB (Delorme & Makeig, 2004). The 1,000 ms from the trial onset to the go-stimulus onset was used as baseline. The number of cycles in Morlet wavelet was set to be 4, which produced ERSPs from about 7 Hz, allowing us to test for mu ERD with the consideration of the trade-off between time and frequency resolutions.

As was discussed, motor-related mu rhythms are most prominent from sites over the sensorimotor cortex and have larger values from sites contralateral to the responding hand. To quantify this lateralization, time–frequency domain LRP-type measures were calculated (De Jong et al., 2006; Deiber et al., 2012; Doyle et al., 2005; Gladwin et al., 2008). Specifically, for each participant and each trial, time–frequency decomposition and ERSP calculation were performed for EEG signals from C3 minus C4 or C4 minus C3, depending on the hand not-to-be-stopped; and then averaged across trials, separately for signal-inhibit and signal-respond trials.

⁴ Specifically, for each participant and separately for signal-inhibit and signal-respond trials, the LRP was the average of the followings: for trials with the X shown on the left side, EEGs from the C4 site were subtracted from EEGs from the C3 site. For trials with the X shown on the right side, EEGs from the C3 site were subtracted from EEGs from the C4 site. Then the average of the LRP across all participants was obtained, separately for signal-inhibit and signal-respond trials.

Results

The practice block was excluded from all analyses. Except for stop-signal trials with stopping failure (i.e. signal-respond trials), trials with other forms of error (e.g. index versus middle fingers or responses with more than two fingers, etc.) were excluded (3.04 % of total trials). Trials with RTs faster than 100 ms or slower than 1,000 ms were excluded (0.4 % of correct trials). Also excluded from analyses were occasional controlled-go trials in which participants made correct responses *before* the appearance of the X (0.19 % of correct trials). Blinks and horizontal eye movements were excluded (3.88 and 1.76 % of correct trials, respectively).⁵

Behavioral analyses and RTs

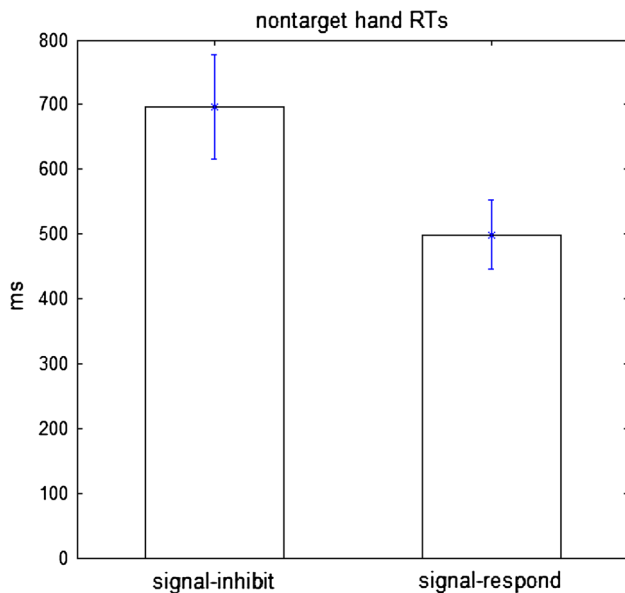
Table 1 shows the probability of stopping, SSD, SSRT, and the nontarget hand RTs in the conditions of interest.⁶ The probability of successful stopping was approximately fifty percent, indicating that the tracking method of SSD worked well. The nontarget hand RTs were shorter in signal-respond than in no-signal-go trials [$F(1,16) = 64.94$, $p < .0001$] or controlled-go trials [$F(1,16) = 174.48$, $p < .0001$], consistent with the independence assumption

⁵ Rejecting trials with blinks: for each trial, the difference between the maximum and the minimum values of vertical EOG, referred to here as peak-to-peak amplitude or PPA, was obtained from the interval between the trial onset (the presentation of the four outline squares) and the RT of that trial. For each participant, the distribution of PPA values from all trials was then obtained. Blinks were assumed to occur in trials with large and deviate PPA values, and those trials were therefore excluded. Rejecting trials with horizontal eye movements: for each trial, the maximum difference between the average of baseline horizontal EOG values and the horizontal EOG values from the interval between the trial onset and the RT of that trial, referred to here as maximum deviate from baseline or MDFB, was obtained. For each participant, the distribution of MDFB values from all trials was then obtained. Horizontal eye movements were assumed to occur in trials with large and deviate MDFB values, and those trials were therefore excluded. A description about the rationale of using PPA and MDFB can be found in Chapter 10 of Jeff Miller's documentation named "ERP software documentation". After the exclusions, the average and the minimum numbers of signal-inhibit trials across all participants were 37.4 (94.7 % of all signal-inhibit trials) and 23, respectively. The average and the minimum numbers of signal-respond trials across all participants were 36.6 (95.5 % of all signal-respond trials) and 15, respectively. The average and the minimum numbers of controlled-go trials across all participants were 72.4 (95.0 % of all controlled-go trials) and 46, respectively. The average and the minimum numbers of no-signal-go trials across all participants were 294 (95.0 % of all no-signal-go trials) and 198, respectively.

⁶ For signal-inhibit and signal-respond trials (i.e. stop-signal trials), the nontarget hand was the hand not-to-be-stopped (i.e. the hand on the contralateral side of the X indicating stopping); for controlled-go trials, the nontarget hand referred to the hand on the contralateral side of the X used to control for the mere presence of the stop signal.

Table 1 Probability of stopping, SSD, SSRT, and the nontarget hand RTs in conditions of interest

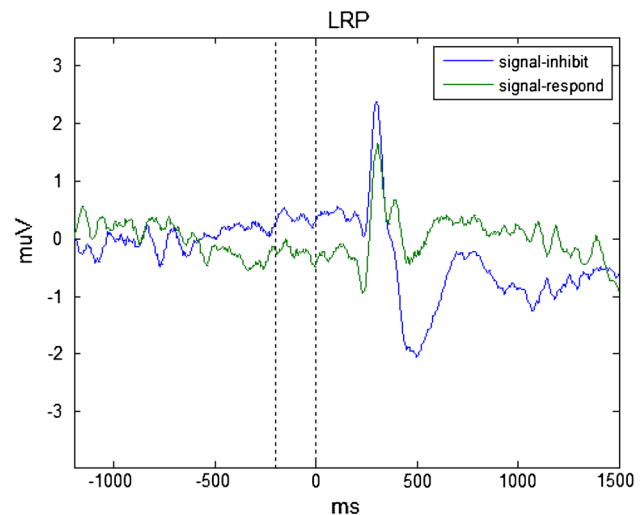
	Grand average	Standard deviation based on between-subject variance
Probability of stopping	50.75 %	0.036
SSD	273 ms	68
No-signal-go RT	544 ms	61
Controlled-go RT	606 ms	71
Signal-respond RT	499 ms	55
Signal-inhibit RT	696 ms	79
SSRT (based on no-signal-go RT)	271 ms	27
SSRT (based on controlled-go RT)	333 ms	29

**Fig. 2** The nontarget hand RTs in signal-inhibit and signal-respond trials

of the race model (Logan & Cowan, 1984; Verbruggen & Logan, 2009a). More importantly, the present task allowed us to assess RTs even when stopping was successful. In line with a previous finding (Ko & Miller, 2011), we demonstrated that the nontarget hand RT was shorter in signal-respond than in signal-inhibit trials [see Fig. 2; $F(1,16) = 215, p < .0001$], as was predicted by the race models.

The LRP

Averages of the LRP time locked to the stop-signal onset in signal-inhibit and signal-respond trials are shown in Fig. 3. To assess influences of voluntarily-generated unimanual preparation on stopping success, we compared the LRPs in

**Fig. 3** LRPs time locked to the stop-signal onset in signal-inhibit and signal-respond trials. The zero point on the horizontal axis is the stop-signal onset

the 200-ms interval before the stop-signal onset between signal-inhibit and signal-respond trials. We found that the pre-signal LRPs were different between the two types of trials [$F(1,16) = 4.60, p < .05$], suggesting that voluntarily-generated unimanual preparation affects stopping success.⁷

It is possible that participants could try to guess the side of the stop signal before its actual presentation and shift their attention to that side beforehand, and therefore contributed to the difference of the pre-signal LRPs between signal-inhibit and signal-respond trials.⁸ To assess this possibility of an attentional effect, we calculated an LRP-type measure from O1 and O2 sites, referred to here as O1/O2 LRP. We checked the 200-ms time interval right before the stop-signal onset, and found that the O1/O2 LRPs in the interval were not different between signal-inhibit and signal-respond trials [$F(1,16) = 0.76, p > .1$], suggesting that there was no attentional effect in the pre-signal LRP difference between signal-inhibit and signal-respond trials.

The lateralized mu ERD

In addition to the LRP, in this section, we tested influences of voluntarily unimanual response preparation on stopping success using the lateralized mu ERD. The difference of the grand average of ERSP across all participants from signal-inhibit minus signal-respond trials is shown in

⁷ The same pattern of results (i.e. different pre-signal LRPs in signal-inhibit and signal-respond trials) was found when the time interval between 500 and 200 ms before the stop-signal onset was assessed [$F(1,16) = 6.46, p < .05$]; whereas there was no difference between the two types of trials in the time interval between 1,000 and 500 ms before the stop-signal onset [$F(1,16) = 2.94, p > .1$].

⁸ We thank Hartmut Leuthold for pointing out this possibility.

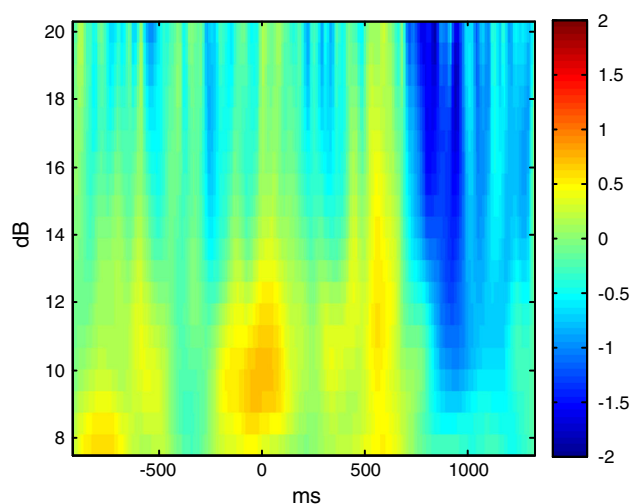


Fig. 4 Lateralized ERD differences calculated by subtracting signal-respond trials from signal-inhibit trials (see text for detail), displaying the frequency range of about 7–21 Hz. The zero point on the horizontal axis is the stop-signal onset

Fig. 4, displaying the frequency range of about 7–21 Hz. As can be seen in Fig. 4, the ERSP difference around the stop-signal onset in the frequency range of about 9–12 Hz looks non-zero.

The average of ERSPs in the frequency range of 9–12 Hz was then calculated. Averages of those ERSPs were shown in Fig. 5, separately for signal-inhibit and signal-respond trials. We can see from Fig. 5 that those ERSPs around the stop-signal onset (time zero) are negative, suggesting they are ERDs. To test influences of voluntarily-generated unimanual preparation, we examined those ERSPs in the 200-ms interval right before the stop-signal onset. The average of those ERSPs in the interval in signal-inhibit trials was significantly smaller than zero (-0.70 dB; $F(1,16) = 6.23$, $p < .05$), indicating lateralized mu ERD in the interval and therefore voluntarily-generated unimanual preparation in signal-inhibit trials. Similarly, the average of those ERSPs in the interval in signal-respond trials was also significantly smaller than zero (-1.27 dB; $F(1,16) = 27.66$, $p < .0001$), indicating lateralized mu ERD in the interval and therefore voluntarily-generated unimanual preparation in signal-respond trials. More importantly, lateralized mu ERD in the interval was more negative in signal-respond than in signal-inhibit trials [$F(1,16) = 5.88$, $p < .05$], indicating larger voluntarily-generated unimanual preparation in signal-respond trials than in signal-inhibit trials. This result clearly suggests that larger voluntarily-generated unimanual preparation contributes to stopping failure.

The controlled-go trials included in the task enabled us to further test whether the difference in voluntarily-

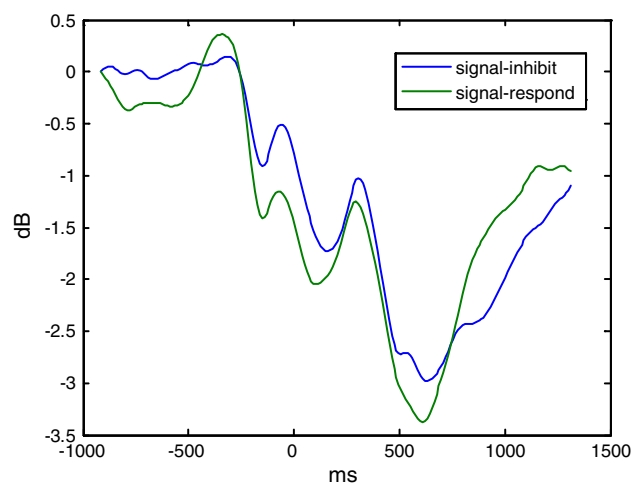


Fig. 5 Lateralized mu ERDs (9–12 Hz), separately for signal-inhibit and signal-respond trials. The zero point on the horizontal axis is the stop-signal onset

generated unimanual preparation between signal-inhibit and signal-respond trials resulted from larger unimanual preparation in signal-respond trials than in most others or from smaller unimanual preparation in signal-inhibit trials than in most others. We therefore compared signal-inhibit and signal-respond trials with controlled-go trials. Lateralized mu ERD in the 200-ms interval right before the stop-signal onset was more negative in signal-respond than in controlled-go trials [-1.27 vs. -0.72 dB; $F(1,16) = 7.69$, $p < .05$]. In contrast, lateralized mu ERDs in the interval were not different between signal-inhibit and controlled-go trials [$F(1,16) = 0.002$, $p > .1$]. These results suggest that voluntarily-generated unimanual preparation that is larger than what is normally produced contributes to stopping failure.

Discussion

Stopping success (signal inhibit vs. signal respond) is attributed to the finishing times of a go and a stop process, according to the race models of the stop-signal paradigm (Boucher et al., 2007; Logan & Cowan, 1984; Verbruggen & Logan, 2009a). In addition to those factors involving processing times, in the present study we sought to use electrophysiological measures to find factors involving activations that could affect stopping success.

Previous two-choice RT studies without using any precue strongly suggest that voluntarily-generated unimanual preparation affects RTs and response accuracy (e.g. Gratton et al., 1988). However, whether voluntarily-generated unimanual preparation plays a role in stopping success is not clear, because most previous studies concerning

prestimulus preparation in the stop-signal paradigm used precues to manipulate prestimulus preparation (e.g. Cai et al., 2011; Claffey et al., 2010; Jentzsch et al., 2004). Therefore, the prestimulus preparation in stop-signal tasks previously studied was not voluntarily generated. The race models of the stop-signal paradigm are based on tasks without any precue, however.

To assess voluntarily-generated unimanual preparation in stop-signal tasks that is conceptually similar to prestimulus unimanual preparation in typical two-choice RT tasks without any precue, we assessed unimanual preparation before the lateralized stop signal (indicating which hand to stop) in a selective-stopping task without any precue. The selective-stopping task that we used also allowed us to assess RTs even when stopping was successful.

In line with a previous finding of Ko and Miller (2011), we found shorter RTs in signal-respond than in signal-inhibit trials. More importantly, two electrophysiological measures of unimanual preparation were used, and comparisons were made between signal-inhibit and signal-respond trials in a time interval right before the lateralized stop signal. The first measure that we used was the LRP, because the LRP has been routinely used to indicate unimanual preparation. We found that pre-signal LRPs were different between the two types of trials, suggesting that voluntarily-generated unimanual preparation affects stopping performance. The second measure that we used was the lateralized mu ERD, because previous studies have found lateralized mu ERD before voluntary movement. We found that pre-signal lateralized mu ERD was more negative in signal-respond than in signal-inhibit trials, indicating that voluntarily-generated unimanual preparation was larger in signal-respond than in signal-inhibit trials. Further analyses with controlled-go trials indicated that voluntarily-generated unimanual preparation in signal-respond trials was larger than that in most other trials, suggesting that larger voluntarily-generated unimanual preparation contributes to unsuccessful stopping rather than smaller voluntarily-generated unimanual preparation contributes to successful stopping.

In sum, in addition to those factors involving finishing times described by the race models, the present findings demonstrated that a factor involving activations, namely voluntarily-generated unimanual preparation, could affect stopping success.

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